

Jake's Daily News Review

Project Documentation & Reference Guide

Automated AI-Powered News Briefing System
Last Updated: March 2026

1. Project Overview

Jake's Daily News Review is a fully automated, cloud-based news briefing system that runs every morning without any manual intervention. It scrapes RSS feeds from reputable sources across four topic categories, summarises each article using a free AI language model, and delivers a beautifully formatted HTML email digest at zero ongoing cost.

The system requires no local hardware — it executes entirely on GitHub's free cloud servers on a daily schedule.

How It Works

The pipeline runs in five stages:

1. GitHub Actions triggers the script automatically each morning at 7:00 AM EDT (11:00 AM UTC).
2. feedparser pulls the latest articles from RSS feeds across all four categories.
3. trafilatura extracts full article text; falls back to the RSS summary if scraping fails.
4. Each article is sent to Groq's free Llama 3.3 70B API, which returns a 2-3 sentence summary.
5. All summaries are compiled into a styled HTML email and delivered via Brevo SMTP.

2. Associated Accounts

The project relies on three free accounts. All credentials are stored as encrypted GitHub Secrets — never visible in code or logs.

GitHub

Field	Value
URL	https://github.com
Purpose	Hosts the code and runs the workflow on schedule
Repository	Daily-Mailer (private)
Workflow file	.github/workflows/daily_brief.yml
Secrets location	Settings > Secrets and variables > Actions > Repository secrets

Groq

Field	Value
URL	https://console.groq.com

Purpose	Free LLM API — summarises articles using Llama 3.3 70B
Model	llama-3.3-70b-versatile
Free limit	~14,400 requests/day
Secret name	GROQ_API_KEY
Key format	Starts with gsk_

Brevo

Field	Value
URL	https://www.brevo.com
Purpose	Free SMTP email delivery — sends to anyone with no restrictions
SMTP host	smtp-relay.brevo.com
SMTP port	587
SMTP login	Found on Brevo > SMTP & API > SMTP tab (e.g. a54321001@smtp-brevo.com)
Free limit	300 emails/day, 9,000/month
Secret name	BREVO_SMTP_KEY + BREVO_SMTP_LOGIN

Brevo Signup Steps

1. Go to <https://www.brevo.com> and click Sign Up Free.
2. Enter your email address and create a password.
3. Check your inbox and click the verification link.
4. Fill in your name and company — use anything, e.g. "Jake" and "Personal".
5. When asked about use case, select Transactional emails.
6. If asked for a physical address, enter any real-looking address.
7. Inside the dashboard go to your account name (top right) > SMTP & API.
8. Click the SMTP tab — note the Login value (e.g. a54321001@smtp-brevo.com).
9. Click Generate a new SMTP Key — copy it immediately.
10. Save the Login as BREVO_SMTP_LOGIN and the key as BREVO_SMTP_KEY in GitHub secrets.

Recipients

Field	Value
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Secret name	EMAIL_RECIPIENT
Format	Comma-separated email addresses
Example	you@gmail.com, friend@gmail.com, colleague@outlook.com
Privacy	All recipients are BCC'd — addresses hidden from each other
Restriction	None — Brevo sends to any valid email address freely
To add recipients	Update EMAIL_RECIPIENT secret — no code changes needed

3. Repository Structure

File	Location	Purpose
daily_brief.py	/ (root)	Main Python script — fetch, summarise, send
requirements.txt	/ (root)	Python dependencies: feedparser, trafilatura, groq
daily_brief.yml	.github/workflows/	GitHub Actions workflow — schedule and secrets

4. GitHub Secrets

Navigate to: Your repo > Settings > Secrets and variables > Actions > Repository secrets

Secret Name	Value	Where to Get It
BREVO_SMTP_LOGIN	e.g. a54321001@smtp-brevo.com	Brevo > SMTP & API > SMTP tab > Login field
BREVO_SMTP_KEY	Generated key from Brevo	Brevo > SMTP & API > SMTP tab > Generate new key
EMAIL_SENDER	your@email.com	The email address you signed up to Brevo with
EMAIL_RECIPIENT	you@gmail.com, ...	Any email addresses, comma separated
GROQ_API_KEY	gsk_XXXXXXXXXXXX	console.groq.com > API Keys > Create API Key

5. News Categories & Feeds

Category	Active Sources
World News	BBC World, Reuters, AP News, Al Jazeera, Deutsche Welle
Tech / AI	TechCrunch, The Verge, Hacker News, Ars Technica, MIT Tech Review

Finance / Markets	MarketWatch, WSJ Markets, The Economist
Science	Science Daily, Nature, Quanta Magazine, Phys.org

To add feeds, edit the FEEDS dictionary in `daily_brief.py`. Most major news sites publish RSS feeds — search "[site name] RSS feed" to find them.

6. Configuration Settings

Setting	Default	Description
groq_model	llama-3.3-70b-versatile	LLM model used for summarisation
articles_per_feed	3	Number of articles pulled per RSS source
recency_hours	24	Only include articles from the last N hours

7. Schedule

The workflow runs on a cron schedule in `daily_brief.yml`. Currently set to 7:00 AM EDT (11:00 AM UTC) daily.

Desired Time	Cron Value
7:00 AM UTC	0 7 * * *
7:00 AM UK / GMT	0 7 * * *
7:00 AM UK / BST (summer)	0 6 * * *
7:00 AM EDT (UTC-4) — current setting	0 11 * * *
7:00 AM EST (UTC-5)	0 12 * * *

Use <https://crontab.guru> to build custom schedules. GitHub may delay scheduled runs by up to 15 minutes.

8. Troubleshooting Guide

How to View Error Logs

1. Go to your GitHub repo and click the Actions tab.
2. Click the most recent failed run.
3. Click send-brief to expand the job.
4. Scroll to the bottom of the red step to find the full traceback.

Common Errors

Error	Cause	Fix
SMTPAuthenticationError (535)	Wrong SMTP login — using account email instead of Brevo SMTP login	Set BREVO_SMTP_LOGIN to the value shown on Brevo > SMTP & API
KeyError: BREVO_SMTP_LOGIN	Secret missing from GitHub or workflow	Add BREVO_SMTP_LOGIN to GitHub secrets. Confirm daily_brief.yml
Emails not arriving	EMAIL_RECIPIENT has a typo or wrong address	Update EMAIL_RECIPIENT secret. Use comma-separated addresses. Brevo
Groq 400: Model decommissioned	Groq model ID no longer supported	Update groq_model to llama-3.3-70b-versatile. Check console.groq.com/
Finance: No content to summarise	Finance sites block scrapers	Already fixed — fallback chain uses RSS summary then title.
Workflow not triggering	GitHub delays up to 15 minutes on schedule	Trigger manually via Actions > Daily Brief > Run workflow.
0 articles for a category	RSS feed URL has changed or is down	Paste the URL into a browser to verify. Replace broken URLs in FEEDS
ModuleNotFoundError	Python dependency missing	Check requirements.txt has: feedparser, trafilatura, groq.
KeyError: secret name	GitHub Secret missing or misspelled	Verify all 5 secrets exist with exact names in Settings > Secrets > Actions
Node.js deprecation warning	Older Node.js on Actions runner	Set FORCE_JAVASCRIPT_ACTIONS_TO_NODE24: true in workflow

Verifying Secret Values

GitHub Secrets cannot be read back once saved. To check temporarily, add this to main() then remove after one run:

```
print(os.environ.get('SECRET_NAME', 'NOT SET'))
```

Regenerating Credentials

Service	Steps
Brevo SMTP Key	brevo.com > SMTP & API > SMTP tab > Generate new key > Update BREVO_SMTP_KEY secret
Groq API Key	console.groq.com > API Keys > Delete old > Create new > Update GROQ_API_KEY secret

9. Adding Recipients

To add people, update EMAIL_RECIPIENT in GitHub Settings > Secrets > Actions:

1. Click the pencil icon next to EMAIL_RECIPIENT.
2. Enter comma-separated addresses: you@gmail.com, friend@gmail.com
3. Click Save — no code changes required.
4. All recipients are automatically BCC'd so no one can see each other's addresses.

10. Extending the Project

Adding New RSS Feeds

Example Source	RSS URL
Guardian World	https://www.theguardian.com/world/rss
Bloomberg Tech	https://feeds.bloomberg.com/technology/news.rss
BBC Science	https://feeds.bbc.co.uk/news/science_and_environment/rss.xml
FT Markets	https://www.ft.com/rss/home/uk

Changing the AI Model

Model ID	Quality	Speed
llama-3.3-70b-versatile	Best — recommended	Medium
llama-3.1-8b-instant	Good	Fastest
qwen/qwen3-32b	Very good	Fast
gemma2-9b-it	Good	Fast

Always verify current model IDs at console.groq.com/docs/models as Groq periodically decommissions old models.